



UNIVERSITI
TEKNOLOGI
PETRONAS

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Student ID: 22007447

Program: Bachelor of Computer Science

Course: Data Visualization (TEB3133)

Lab 6

Module 1 & 2

[illegible]

Module 3

The screenshot shows the Power BI Desktop interface. On the left, the 'Queries' pane lists five queries: 'FinanceTable', 'DimType', 'DimDate', 'DimComponent', and 'FactFinance'. The 'DimType' query is selected. On the right, the 'Table' view displays a table with two columns: 'A^B C Type' and '1^2_3 TypeKey'. The table contains four rows of data:

A^B C Type	1^2_3 TypeKey
1 Income	1
2 Savings	2
3 Expense	3
4 Target	4

Queries [5] <

FinanceTable
DimType
DimDate
DimComponent
FactFinance

= Table.RenameColumns(#"Added Index",{{"Index", "ComponentKey"}})

	^B _C Type	^B _C Component	^B _C Date	¹ ₂ Amount	¹ ₃ ComponentKey
1	Income	Salary	1/1/2023	60000	1
2	Income	Freelancig	1/1/2023	1000	2
3	Savings	Mutual funds	1/1/2023	6000	3
4	Savings	Emergency Fund	1/1/2023	2000	4
5	Savings	Fixed Deposit	1/1/2023	2000	5
6	Savings	Liquid Cash	1/1/2023	35600	6
7	Expense	House Rent	1/1/2023	10000	7
8	Expense	Groceries & Food	1/1/2023	6000	8
9	Expense	Health	1/1/2023	1000	9
10	Expense	EMIs	1/1/2023	2600	10
11	Expense	Leisure	1/1/2023	600	11
12	Expense	Shopping	1/1/2023	1000	12
13	Expense	Trevel	1/1/2023	200	13
14	Expense	Other	1/1/2023	100	14
15	Target	Savings	1/1/2023	0.25	15

Queries [5] <

FinanceTable
DimType
DimDate
DimComponent
FactFinance

= Table.Distinct(#"Removed Other Columns1")

	^B _C Date
1	1/1/2023
2	1/2/2023
3	1/3/2023
4	1/4/2023
5	1/5/2023
6	1/6/2023
7	1/7/2023
8	1/8/2023
9	1/9/2023
10	1/10/2023
11	1/11/2023
12	1/12/2023
13	1/1/2024
14	1/2/2024
15	1/3/2024
16	1/4/2024
17	1/5/2024
18	1/6/2024
19	1/7/2024
20	1/8/2024
21	1/9/2024
22	1/10/2024
23	1/11/2024
24	1/12/2024
25	1/1/2025
26	1/2/2025
27	1/3/2025
28	1/4/2025
29	1/5/2025
30	1/6/2025
31	1/7/2025
32	1/8/2025
33	1/9/2025
34	1/10/2025
35	1/11/2025

Custom Column

Add a column that is computed from the other columns.

New column name

DateKey

Custom column formula ⓘ

= Date.Year([Date]) * 100 + Date.Month([Date])

Available columns

Date

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK

Cancel

fx

= Table.AddColumn(#"Changed Type2", "DateKey", each Date.Year([Date]) * 100 + Date.Month([Date])

	Date	DateKey
1	1/1/2023	202301
2	1/2/2023	202302
3	1/3/2023	202303
4	1/4/2023	202304
5	1/5/2023	202305
6	1/6/2023	202306
7	1/7/2023	202307
8	1/8/2023	202308
9	1/9/2023	202309
10	1/10/2023	202310
11	1/11/2023	202311
12	1/12/2023	202312
13	1/1/2024	202401
14	1/2/2024	202402
15	1/3/2024	202403
16	1/4/2024	202404
17	1/5/2024	202405
18	1/6/2024	202406
19	1/7/2024	202407
20	1/8/2024	202408
21	1/9/2024	202409
22	1/10/2024	202410
23	1/11/2024	202411
24	1/12/2024	202412
25	1/1/2025	202501
26	1/2/2025	202502
27	1/3/2025	202503
28	1/4/2025	202504
29	1/5/2025	202505
30	1/6/2025	202506
31	1/7/2025	202507
32	1/8/2025	202508
33	1/9/2025	202509
34	1/10/2025	202510

Query Settings

PROPERTIES

Name

DimDate

All Properties

APPLIED STEPS

Source

Navigation

Changed Type

Promoted Headers

Changed Type1

Unpivoted Other Columns

Renamed Columns1

Removed Other Columns1

Removed Duplicates1

Changed Type2

Added Custom

Unpivoted Columns

Removed Columns

Renamed Columns

Removed Other Columns

Removed Duplicates

Queries [5] = Table.AddColumn(#"Added Custom3", "Year", each Date.Year([Date]))

	ABC 123	ABC 123	ABC 123	ABC 123	ABC 123
	Date	DateKey	Month	MonthNumber	Quarter
1	1/1/2023	202301	January	1	Q1
2	1/2/2023	202302	February	2	Q1
3	1/3/2023	202303	March	3	Q1
4	1/4/2023	202304	April	4	Q2
5	1/5/2023	202305	May	5	Q2
6	1/6/2023	202306	June	6	Q2
7	1/7/2023	202307	July	7	Q3
8	1/8/2023	202308	August	8	Q3
9	1/9/2023	202309	September	9	Q3
10	1/10/2023	202310	October	10	Q4
11	1/11/2023	202311	November	11	Q4
12	1/12/2023	202312	December	12	Q4
13	1/1/2024	202401	January	1	Q1
14	1/2/2024	202402	February	2	Q1
15	1/3/2024	202403	March	3	Q1
16	1/4/2024	202404	April	4	Q2
17	1/5/2024	202405	May	5	Q2
18	1/6/2024	202406	June	6	Q2
19	1/7/2024	202407	July	7	Q3
20	1/8/2024	202408	August	8	Q3
21	1/9/2024	202409	September	9	Q3
22	1/10/2024	202410	October	10	Q4
23	1/11/2024	202411	November	11	Q4
24	1/12/2024	202412	December	12	Q4
25	1/1/2025	202501	January	1	Q1
26	1/2/2025	202502	February	2	Q1
27	1/3/2025	202503	March	3	Q1
28	1/4/2025	202504	April	4	Q2
29	1/5/2025	202505	May	5	Q2
30	1/6/2025	202506	June	6	Q2
31	1/7/2025	202507	July	7	Q3
32	1/8/2025	202508	August	8	Q3
33	1/9/2025	202509	September	9	Q3
34	1/10/2025	202510	October	10	Q4
35	1/11/2025	202511	November	11	Q4
36	1/12/2025	202512	December	12	Q4

Queries [5] = Table.ExpandTableColumn(#"Merged Queries", "DimDate", {"DateKey"}, {"DateKey"})

	ABC 123	ABC 123	123	123	ABC 123
	Component	Date	Amount	FinancelD	DateKey
1	Salary	1/1/2023	60000	1	202301
2	Salary	1/1/2023	60000	2	202301
3	Salary	1/1/2023	60000	3	202301
4	Salary	1/1/2023	60000	4	202301
5	Salary	1/1/2023	60000	5	202301
6	Salary	1/1/2023	60000	6	202301
7	Salary	1/1/2023	60000	7	202301
8	Salary	1/1/2023	60000	8	202301
9	Salary	1/1/2023	60000	9	202301
10	Salary	1/1/2023	60000	10	202301
11	Salary	1/1/2023	60000	11	202301
12	Salary	1/1/2023	60000	12	202301
13	Salary	1/1/2023	60000	13	202301
14	Salary	1/1/2023	60000	14	202301
15	Salary	1/1/2023	60000	15	202301
16	Salary	1/1/2023	60000	16	202301
17	Salary	1/1/2023	60000	17	202301
18	Salary	1/1/2023	60000	18	202301
19	Salary	1/1/2023	60000	19	202301
20	Salary	1/1/2023	60000	20	202301
21	Salary	1/1/2023	60000	21	202301
22	Salary	1/1/2023	60000	22	202301
23	Salary	1/1/2023	60000	23	202301
24	Salary	1/1/2023	60000	24	202301
25	Salary	1/1/2023	60000	25	202301
26	Salary	1/1/2023	60000	26	202301
27	Salary	1/1/2023	60000	27	202301
28	Salary	1/1/2023	60000	28	202301
29	Salary	1/1/2023	60000	29	202301
30	Salary	1/1/2023	60000	30	202301
31	Salary	1/1/2023	60000	31	202301
32	Salary	1/1/2023	60000	32	202301
33	Salary	1/1/2023	60000	33	202301
34	Salary	1/1/2023	60000	34	202301
35	Salary	1/1/2023	60000	35	202301
36	Salary	1/1/2023	60000	36	202301

Merged DimDate and FactFinance

FinanceTable DimType DimDate DimComponent FactFinance	<table><tr><th> </th><th> Date</th><th> 1.2 Amount</th><th> 1²₃ FinanceID</th><th> ABC 123 DateKey</th><th> 1²₃ TypeKey</th></tr><tr><td>1</td><td>1/1/2023</td><td>60000</td><td>1</td><td>202301</td><td>1</td></tr><tr><td>2</td><td>1/1/2023</td><td>60000</td><td>2</td><td>202301</td><td>1</td></tr><tr><td>3</td><td>1/1/2023</td><td>60000</td><td>3</td><td>202301</td><td>1</td></tr><tr><td>4</td><td>1/1/2023</td><td>60000</td><td>4</td><td>202301</td><td>1</td></tr><tr><td>5</td><td>1/1/2023</td><td>60000</td><td>5</td><td>202301</td><td>1</td></tr><tr><td>6</td><td>1/1/2023</td><td>60000</td><td>6</td><td>202301</td><td>1</td></tr><tr><td>7</td><td>1/1/2023</td><td>60000</td><td>7</td><td>202301</td><td>1</td></tr><tr><td>8</td><td>1/1/2023</td><td>60000</td><td>8</td><td>202301</td><td>1</td></tr><tr><td>9</td><td>1/1/2023</td><td>60000</td><td>9</td><td>202301</td><td>1</td></tr><tr><td>10</td><td>1/1/2023</td><td>60000</td><td>10</td><td>202301</td><td>1</td></tr><tr><td>11</td><td>1/1/2023</td><td>60000</td><td>11</td><td>202301</td><td>1</td></tr><tr><td>12</td><td>1/1/2023</td><td>60000</td><td>12</td><td>202301</td><td>1</td></tr><tr><td>13</td><td>1/1/2023</td><td>60000</td><td>13</td><td>202301</td><td>1</td></tr><tr><td>14</td><td>1/1/2023</td><td>60000</td><td>14</td><td>202301</td><td>1</td></tr><tr><td>15</td><td>1/1/2023</td><td>60000</td><td>15</td><td>202301</td><td>1</td></tr><tr><td>16</td><td>1/1/2023</td><td>60000</td><td>16</td><td>202301</td><td>1</td></tr><tr><td>17</td><td>1/1/2023</td><td>60000</td><td>17</td><td>202301</td><td>1</td></tr><tr><td>18</td><td>1/1/2023</td><td>60000</td><td>18</td><td>202301</td><td>1</td></tr><tr><td>19</td><td>1/1/2023</td><td>60000</td><td>19</td><td>202301</td><td>1</td></tr><tr><td>20</td><td>1/1/2023</td><td>60000</td><td>20</td><td>202301</td><td>1</td></tr><tr><td>21</td><td>1/1/2023</td><td>60000</td><td>21</td><td>202301</td><td>1</td></tr><tr><td>22</td><td>1/1/2023</td><td>60000</td><td>22</td><td>202301</td><td>1</td></tr><tr><td>23</td><td>1/1/2023</td><td>60000</td><td>23</td><td>202301</td><td>1</td></tr><tr><td>24</td><td>1/1/2023</td><td>60000</td><td>24</td><td>202301</td><td>1</td></tr><tr><td>25</td><td>1/1/2023</td><td>60000</td><td>25</td><td>202301</td><td>1</td></tr><tr><td>26</td><td>1/1/2023</td><td>60000</td><td>26</td><td>202301</td><td>1</td></tr><tr><td>27</td><td>1/1/2023</td><td>60000</td><td>27</td><td>202301</td><td>1</td></tr><tr><td>28</td><td>1/1/2023</td><td>60000</td><td>28</td><td>202301</td><td>1</td></tr><tr><td>29</td><td>1/1/2023</td><td>60000</td><td>29</td><td>202301</td><td>1</td></tr><tr><td>30</td><td>1/1/2023</td><td>60000</td><td>30</td><td>202301</td><td>1</td></tr><tr><td>31</td><td>1/1/2023</td><td>60000</td><td>31</td><td>202301</td><td>1</td></tr><tr><td>32</td><td>1/1/2023</td><td>60000</td><td>32</td><td>202301</td><td>1</td></tr><tr><td>33</td><td>1/1/2023</td><td>60000</td><td>33</td><td>202301</td><td>1</td></tr><tr><td>34</td><td>1/1/2023</td><td>60000</td><td>34</td><td>202301</td><td>1</td></tr><tr><td>35</td><td>1/1/2023</td><td>60000</td><td>35</td><td>202301</td><td>1</td></tr><tr><td>36</td><td>1/1/2023</td><td>60000</td><td>36</td><td>202301</td><td>1</td></tr></table>		Date	1.2 Amount	1 ² ₃ FinanceID	ABC 123 DateKey	1 ² ₃ TypeKey	1	1/1/2023	60000	1	202301	1	2	1/1/2023	60000	2	202301	1	3	1/1/2023	60000	3	202301	1	4	1/1/2023	60000	4	202301	1	5	1/1/2023	60000	5	202301	1	6	1/1/2023	60000	6	202301	1	7	1/1/2023	60000	7	202301	1	8	1/1/2023	60000	8	202301	1	9	1/1/2023	60000	9	202301	1	10	1/1/2023	60000	10	202301	1	11	1/1/2023	60000	11	202301	1	12	1/1/2023	60000	12	202301	1	13	1/1/2023	60000	13	202301	1	14	1/1/2023	60000	14	202301	1	15	1/1/2023	60000	15	202301	1	16	1/1/2023	60000	16	202301	1	17	1/1/2023	60000	17	202301	1	18	1/1/2023	60000	18	202301	1	19	1/1/2023	60000	19	202301	1	20	1/1/2023	60000	20	202301	1	21	1/1/2023	60000	21	202301	1	22	1/1/2023	60000	22	202301	1	23	1/1/2023	60000	23	202301	1	24	1/1/2023	60000	24	202301	1	25	1/1/2023	60000	25	202301	1	26	1/1/2023	60000	26	202301	1	27	1/1/2023	60000	27	202301	1	28	1/1/2023	60000	28	202301	1	29	1/1/2023	60000	29	202301	1	30	1/1/2023	60000	30	202301	1	31	1/1/2023	60000	31	202301	1	32	1/1/2023	60000	32	202301	1	33	1/1/2023	60000	33	202301	1	34	1/1/2023	60000	34	202301	1	35	1/1/2023	60000	35	202301	1	36	1/1/2023	60000	36	202301	1
	Date	1.2 Amount	1 ² ₃ FinanceID	ABC 123 DateKey	1 ² ₃ TypeKey																																																																																																																																																																																																																										
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36	1/1/2023	60000	36	202301	1																																																																																																																																																																																																																										

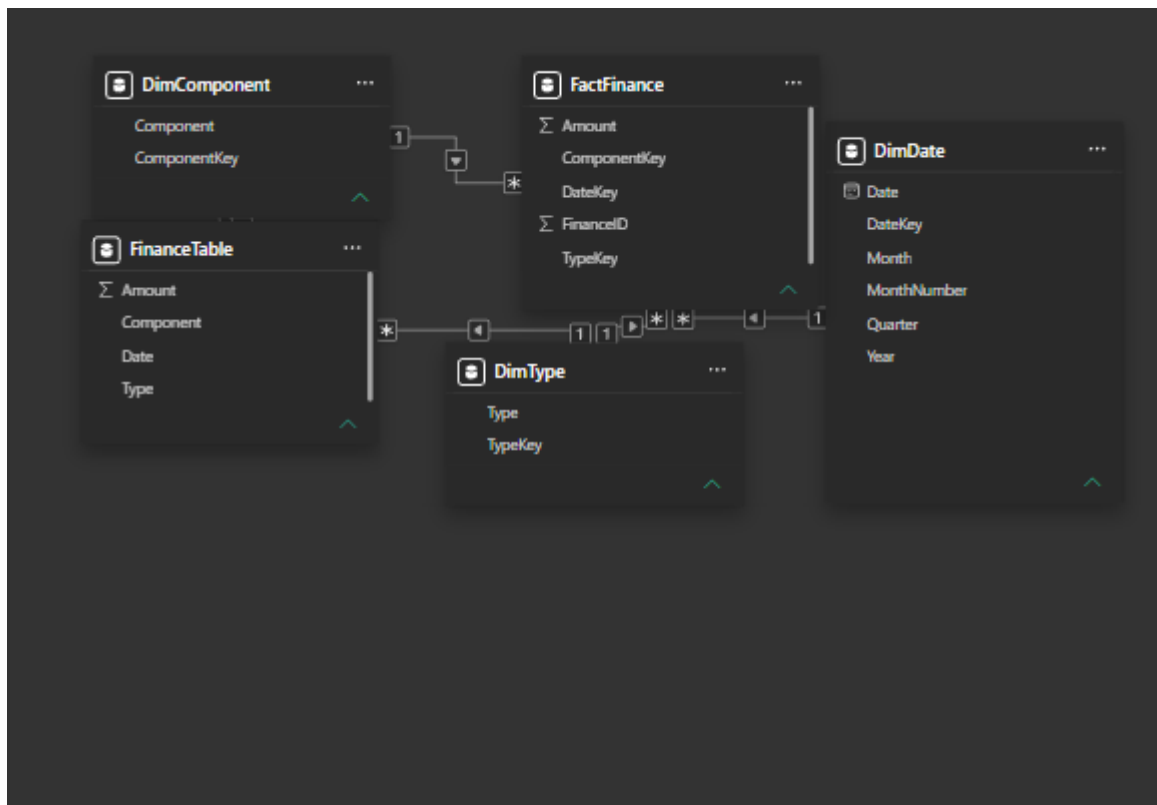
Merged FactFinance and DimType

Queries [5]

Table.ExpandTableColumn("#Merged Queries2", "DimComponent", {"ComponentKey"}, {"ComponentKey"})

	1.2	Amount	1.2.3	FinanceID	ABC 123	DateKey	1.2.3	TypeKey	1.2.3	ComponentKey
1	23	60000			1	202301		1		1
2	23	60000			2	202301		1		1
3	23	60000			3	202301		1		1
4	23	60000			4	202301		1		1
5	23	60000			5	202301		1		1
6	23	60000			6	202301		1		1
7	23	60000			7	202301		1		1
8	23	60000			8	202301		1		1
9	23	60000			9	202301		1		1
10	23	60000			10	202301		1		1
11	23	60000			11	202301		1		1
12	23	60000			12	202301		1		1
13	23	60000			13	202301		1		1
14	23	60000			14	202301		1		1
15	23	60000			15	202301		1		1
16	23	60000			16	202301		1		1
17	23	60000			17	202301		1		1
18	23	60000			18	202301		1		1
19	23	60000			19	202301		1		1
20	23	60000			20	202301		1		1
21	23	60000			21	202301		1		1
22	23	60000			22	202301		1		1
23	23	60000			23	202301		1		1
24	23	60000			24	202301		1		1
25	23	60000			25	202301		1		1
26	23	60000			26	202301		1		1
27	23	60000			27	202301		1		1
28	23	60000			28	202301		1		1
29	23	60000			29	202301		1		1
30	23	60000			30	202301		1		1
31	23	60000			31	202301		1		1
32	23	60000			32	202301		1		1
33	23	60000			33	202301		1		1
34	23	60000			34	202301		1		1
35	23	60000			35	202301		1		1

Merged FactFinance and DimComponent



Finalized Dashboard

5.85M

Total Amount

15

Number of Components

2.79M

Total Income

1.79M

Total Savings

Component

All

Year

All

Month

All

